

Internal Hackathon — 2-Feature Prototype Plan

Features: (1) Susceptibility Model + GIS Heatmap (2) Real-Time Alerts + Citizen Reporting

Team split (6 members, 4 days)

Role	Days 1-2	Days 3-4
A — Data/GIS	Pull DEM + GSI landslide points for chosen state, compute slope/aspect/drainage in QGIS	Finalize GIS layers, hand off to dashboard
B — ML	Feature engineering, train logistic regression / random forest, validate	Wire model output to an API endpoint, write validation summary for slide 5
C — Backend	API + DB schema, rainfall API integration	Alert trigger logic (rainfall threshold → alert fire)
D — Frontend	React + Leaflet skeleton	Heatmap layer wired to live model output, risk panel
E — Reporting	Simple web form for geo-tagged photo upload (skip native app — a mobile-responsive web form is enough for this round)	Wire submissions to dashboard, test end-to-end
F — PM/Pitch	Coordinate scope, own deck (already built — see attached)	Fill in real validation numbers on slide 5, run 2 mock Q&A sessions using the prep sheet below

What “done” looks like by Day 4

- Model trained on real data, one honest validation number ready to say out loud
- Dashboard shows the model’s actual output, not a static mockup

- One alert has fired live during rehearsal, triggered by a real or simulated rainfall value crossing threshold
- One citizen report has flowed from form → backend → dashboard, live

Deck notes

- **Slide 5** has a placeholder `[AUC / accuracy on held-out points]` — replace with your real validation metric before presenting. Do not leave a placeholder or invent a number.
- Slides 3 and 9 are your “honesty” slides — they’re the direct answer to “why only two features” and “what’s real vs. not.” Don’t cut these to save time; they’re doing more Q&A defense work than any other slide.
- Team name/branch on slides 1 and 11 need filling in.

Quick Q&A prep (condensed for this round)

- **“Why only two features?”** → point to slide 3: depth over breadth, rest is a documented roadmap.
- **“Is this really AI?”** → point to slide 4: static ML layer + dynamic rule layer, named separately on purpose.
- **“How accurate is it?”** → give the real number from slide 5, and be clear it’s zone-level susceptibility, not exact-time prediction.
- **“What about soil sensors?”** → slide 9, roadmap column: satellite proxy now, ground sensors are phase 2.